

CDS6314 DATA MINING

FCI-2530

Project

Shopper Patterns and Analytics Research Kit (SPARK)

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Submission Date: 6/2/2025

Abstract

The Shopper Patterns and Analytics Research Kit (SPARK) is a data mining framework designed to analyze online shopping behavior and predict consumer purchasing intent. This study utilizes the UCI Online Shoppers Purchasing Intention Dataset, comprising 12,330 unique user sessions, and adopts a dual-method analytical approach that integrates machine learning classification with association rule mining. Two classification models, namely Logistic Regression and Decision Tree, were developed to predict session-level purchase outcomes. Experimental results show that the Decision Tree model achieved a higher recall score of 0.6466, successfully identifying approximately 50% more purchasing sessions than Logistic Regression. In addition, association rule mining revealed high-lift behavioral patterns, including strong correlations between returning visitors with high page values and successful conversions. To enhance practical applicability, the analytical findings were incorporated into an interactive Streamlit-based dashboard, enabling e-commerce stakeholders to obtain real-time, actionable insights for targeted marketing strategies and website optimization.

Contents

Abstract	2
1 Introduction.....	4
1.1 Background	4
1.2 Problem Statement	4
1.3 Objectives	4
2 Related Works	5
2.1 Selected Research Papers / Applications	5
2.2 Justification of Selected Works.....	5
3 Methodology.....	6
3.1 Overall Framework	6
3.2 Data Preprocessing	8
3.3 Data Mining Techniques.....	10
3.4 Evaluation	14
3.5 Results and Discussion	18
3.6 Application	21
4 Conclusion and Future Work.....	30
4.1 Conclusion	30
4.2 Future Work	31
References	31
Appendix 1 Task Distribution	31

1 Introduction

1.1 Background

In the competitive e-commerce landscape, understanding user interactions is vital for generating revenue. The SPARK project leverages session-level data, including page visits, duration, bounce rates, and traffic sources, collected over a one-year period to capture seasonal shopping variations. By utilizing Google Analytics tracking mechanisms from real-world websites, this study focuses on the domain of online consumer behavior to drive data-informed decision-making.

1.2 Problem Statement

E-commerce platforms face persistent challenges in accurately distinguishing casual browsers from high-intent shoppers in real time. This limitation often results in inefficient allocation of marketing resources and missed conversion opportunities. In addition, session-level data in e-commerce environments are inherently imbalanced, with only about 16% of user visits culminating in a purchase. Such skewed class distributions make it technically difficult to develop predictive models that can reliably identify potential buyers without generating a high rate of false positives, thereby reducing the practical effectiveness of real-time decision support systems.

1.3 Objectives

The primary objectives of the SPARK project are as follows:

1. **Predictive Modeling**

To design, implement, and evaluate machine learning classification models capable of predicting whether an online user session will result in a purchase, with particular emphasis on handling class imbalance in e-commerce data.

2. **Comparative Model Analysis**

To compare the performance of Logistic Regression and Decision Tree models using appropriate evaluation metrics (e.g., recall, precision, and accuracy) in order to determine their effectiveness in identifying high-intent shoppers.

3. **Pattern Discovery**

To apply association rule mining techniques to uncover meaningful and high-impact behavioral patterns that are strongly associated with successful purchase outcomes.

4. **Practical Application and Visualization**

To integrate the predictive and descriptive analytics results into an interactive Streamlit-based dashboard that provides e-commerce stakeholders with actionable insights for targeted marketing and website optimization.

2 Related Works

2.1 Selected Research Papers / Applications

Online shopping has become a major driver of economic growth and an integral part of daily life. Many recent studies have investigated online shoppers' behavior using various machine learning algorithms.

- Abdullah-All-Tanvir, Khandokar, I. A., Islam, A. M., Islam, S., & Shatabda, S. (2023). A gradient boosting classifier for purchase intention prediction of online shoppers. *Heliyon*, 9(4), e15163. <https://doi.org/10.1016/j.heliyon.2023.e15163>
- Satu, M. S., & Islam, S. F. (2023). Modeling online customer purchase intention behavior applying different feature engineering and classification techniques. *Research Square (Research Square)*. <https://doi.org/10.21203/rs.3.rs-3185752/v1>

2.2 Justification of Selected Works

Previous research has extensively explored online consumer behavior using data-driven approaches and machine learning techniques. The UCI Online Shoppers Purchasing Intention dataset has been a common benchmark for these studies, providing rich session-based features to predict purchasing behavior.

Abdullah-All-Tanvir et al. (2023) proposed a purchased intension prediction model using the same dataset comprising 12330 sessions. The study applied feature selection to identify the most discriminative attributes and trained five classifiers, which are Support Vector Machine (SVM), Random Forest, Multilayer Perceptron, Decision Tree and XGBoost. The study use SMOTE to address dataset's class imbalance. The experimental results shows that XGBoost achieved highest performance with an accuracy of 90.65%, while Decision Tree achieve accuracy of 90.54%. The study then provided an explainable analysis to explain model decisions, highlighting PageValues and ExitRates as among the most influential features for predicting purchase intent. This research paper shows the performance of Decision tree on the UCI Online Shoppers Purchasing Intention dataset and takle class imbalance through oversampling.

Another study that investigated online customer purchase intention prediction is Satu and Islam (2023). This study applied multiple feature engineering and classification techniques to the UCI Online Shoppers Purchasing Intention dataset. The authors generated several transformed versions of the dataset using different feature transformation methods, including Min-Max normalization, Z-score standardization, and square root transformation. Similar to Abdullah-All-Tanvir et al. (2023), SMOTE is applied to balance the classes. Feature Selection such as Chi-Squared Feature Selection, Gain Ratio Attribute Evaluation and Best First Search are applied to produce a large matrix of dataset configurations. This study used 14 classifiers, such as Random Forest, Decision Tree, Logistic Model Trees, Simple CART and more, to evaluate across these configurations. The key finding is that Random Forest achieved highest accuracy, which is 92.39%, when Random Forest was applied to the Z-Score transformed dataset with Gain Ratio Attribute Evaluation feature selection. The study showed that data transformation choices strongly affect model performance. This means preprocessing is an important design decision that works together with model selection.

Both studies used the same UCI benchmark dataset adopted in SPARK, making their reported baselines directly relevant for comparison. Instead of testing many models, this project focus on Logistic Regression and Decision Tree.

3 Methodology

3.1 Overall Framework

The framework of this project is structured into 7 sequential stages as shown below.

Stage	Description
Stage 1- Dataset Acquisition	The UCI Online Shoppers Purchasing Intention Dataset (12,330 sessions, 18 attributes) is loaded and its structure is inspected to confirm completeness.
Stage 2- Data Preprocessing	Exploratory data analysis is conducted, followed by duplicate removal, categorical encoding (One-Hot Encoding), and StandardScaler normalization of numerical features.
Stage 3- Classification	The preprocessed data is divided into 80% training sets and 20% testing sets. Stratification is used to ensure both groups maintain the original ratio. Logistic Regression and Decision Tree are trained to predict purchase intent.
Stage 4- Association Rule Mining	Numerical features are discretized into categorical bins and the data is converted to transactional format The Apriori algorithm extracts frequent itemset and association rules.

Stage5- Evaluation	Both classification models are assessed using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Association rules are evaluated on Support, Confidence, and Lift.
Stage 6- Results and Discussion	Classification outcomes and association rule findings are analysed, key insights are extracted, and business implications are discussed.
Stage 7- Application Development	All analytical results are integrated into the SPARK Streamlit dashboard for interactive exploration and real-time prediction.

3.1.1 Dataset description

The dataset in this project is the Online Shoppers Purchasing Intention Dataset sourced from the UCI Machine Learning Repository. The dataset is introduced by Sakar et al.(2019). It has since become a widely adopted benchmark for e-commerce purchase-intent prediction research.

The dataset Volume is shown in the table below.

Property	Value
Total data	12 330
Total Attributes	18
Numerical Attributes	10
Categorical Attributes	8
Revenue Sessions	1 908
Non- Revenue Sessions	10 422
Class Balance Ratio (Revenue: No Revenue)	1:5.4
Missing Value	0

3.1.2 Source and Collection

The data were collected from the real-word e-commerce website by using Google Analytics tracking mechanism for 1 year. Each row of data represents a unique user session so that no single user dominates the dataset. The target variable, which is 'Revenue', is derived directly from whether the session concluded with a completed purchase transaction.

3.1.3 Attribute Description

Attribute	Type	Description
Administrative		
Administrative	Numerical (int)	Number of administrative pages visited during the session.
Administrative_Duration	Numerical (float)	Total time (seconds) spent on administrative pages.
Informational		

Informational	Numerical (int)	Number of informational pages visited during the session.
Informational_Duration	Numerical (float)	Total time (seconds) spent on informational pages.
Product Related		
ProductRelated	Numerical (int)	Number of product-related pages visited during the session.
ProductRelated_Duration	Numerical (float)	Total time (seconds) spent on product-related pages.
Session & Traffic Metrics		
BounceRates	Numerical (float)	Average bounce rate of pages visited (proportion of visitors who leave after one page).
ExitRates	Numerical (float)	Average exit rate of pages visited (proportion of page-views that were the last in the session).
PageValues	Numerical (float)	Average page value of pages visited before an e-commerce transaction.
SpecialDay	Numerical (float)	Closeness of the visiting time to a special day (e.g., Valentine's Day).
Visitor & Temporal		
OperatingSystems	Numerical (int)	Operating system used by the visitor (encoded as integer).
Browser	Numerical (int)	Browser used by the visitor (encoded as integer).
Region	Numerical (int)	Geographic region of the visitor (encoded as integer).
TrafficType	Numerical (int)	Traffic source that directed the visitor to the site (encoded as integer).
VisitorType	Categorical	Type of visitor: Returning_Visitor, New_Visitor, or Other.
Weekend	Boolean	Whether the session occurred on a weekend (True / False).
Target Variable		
Revenue	Boolean	Target variable. True if the session resulted in a purchase; False otherwise.

3.1.4 Class Imbalance

The dataset shows class imbalance. There are only 15.5% of sessions that result in a purchase while 84.5% do not. This is a 1:54 ratio, which indicates that a naive classifier predicting “No Revenue” for every session would already achieve nearly 84.5% accuracy. Therefore, metrics such as Recall and F1-Score become important for fairly assessing model performance in the ‘Revenue’ class.

3.2 Data Preprocessing

Data preprocessing transforms the raw dataset into model-ready formats.

3.2.1 Data Cleaning

125 duplicate rows of data were detected and removed. The dataset was reduced into a total of 12 205 unique data. There are no missing values or negative values. The data types of all columns are correctly typed.

3.2.2 Exploratory Data Analysis (EDA)

EDA was performed to understand feature distributions, and relationships with the target variable before transformation apply. Key findings are as follows:

- PageValues exhibited the strongest predictive signal for revenue generation among all numerical features.
- BounceRates and ExitRates show a strong positive correlation (0.90), suggesting partial redundancy; both correlate negatively with page engagement metrics.
- Duration features display high variance and extreme outliers. For example, ProductRelated_Duration ranges from 0 to 63 974 seconds against a median of only 609 seconds. These outliers are handled by StandardScaler during the scaling step.
- VisitorType: Returning visitors dominate the dataset (85.45 %). New visitors exhibit higher conversion rate (24.93 % vs. 14.09 %).
- Month: November records the highest conversion rate (25.49 %), aligning with seasonal events such as Black Friday. May and November together account for the largest session volumes.
- Weekend sessions show a marginally higher conversion rate (17.45 % vs. 15.08 %)

3.2.3 Encoding of Categorical Variables

Distinct encoding strategies are applied based on variable type. Binary variables, which are Weekend and Revenue, are converted to 0 and 1 integer values as they are inherently binary. Nominal categorical variables like VisitorType and Month are encoded by One-Hot Encoding with drop_first=True to avoid multicollinearity. This produces binary indicator columns for each category while dropping one reference category per variable. Since OperatingSystems, Browser, Region and TrafficType are already stored as integer in dataset; no further encoding is required.

The feature set then expands from 17 input columns to 26 columns after the One-hot expansion.

3.2.4 Feature Scaling

Feature scaling is used to ensure that numbers with large values do not overpower numbers with small values in the model. StandardScaler, which is z-score normalization, was applied to transform each numerical feature to zero mean and unit variance:

$$z = \frac{x - \mu}{\sigma}$$

Scaling is applied only to the original numerical features, such as page counts, time durations, rates and SpecialDay variables. The binary features created by One-Hot Encoding are not scaled as they already have value 0 or 1.

3.3 Data Mining Techniques

3.3.1 Classification Techniques

In this SPARK project, it focuses on supervised learning to predict whether a visitor will generate revenue during shopping sessions. The goal is to predict the categorical target variable called revenue using input features in which revenue is 1 when it is purchase sessions and 0 when it is non-purchase sessions. To predict the binary purchase outcomes and evaluate different predictive approaches, two following classification models are selected which are Logistic Regression and Decision Tree. Both selected models are trained with the same preprocessed dataset to ensure fair evaluation and comparison.

Before training the model and evaluating the performance of the classification models, the dataset is divided into training and testing subsets after doing preprocessing and scaling. The data splitting will be using the `train_test_split` function from the scikit-learn library. The test size is set to 0.2, which means 80:20 splits where 80% of data are used for model training and 20% of data are for testing unseen data to ensure sufficient test samples for evaluation. The `random_state` is set to 42 to the splitting to guarantee that data split is identical for every execution which can ensure reproducibility of results. Additionally, the `stratify` is set to `y` as there are about 15.5% of the sessions resulting in revenue which leads to imbalanced dataset. With this, it ensures that both train and test sets maintain exact class proportion which can prevent the model being evaluated on a biased subset.

3.3.1.1 Logistic Regression

Logistic regression is selected due to its efficiency and is suitable for binary classification problems. The model applied a logistic sigmoid function to a linear combination of input features to predict the probability that a user session will culminate in revenue based on behavioral and session related attributes.

Before training the logistic regression model, it was initialized with `random_state` at 42 to ensure reproducibility followed by `max_iter` limit at 1000 to ensure convergence during optimization. After the model initialized, it was trained using the split standardized

training dataset. Upon completion of the training, the coefficients were analyzed to understand the influence of the features on the probability of purchase sessions. The features with positive coefficients increase the chances of purchase while the negative coefficients reduce the chances.

Based on the result in Figure 4.3.1.1, it shows that PageValues are the strongest positive coefficients at 1.51395, which indicates that sessions with higher value pages are more likely to result in purchase. In contrast, the month of February is the lowest coefficient at -1.1006, which represents the strongest negative impact on purchase probability. Additionally, the model intercept of -1.9920 indicates that the sessions are likely to be non-purchase unless the positive coefficients features are involved.

After training the model, the predictions are generated for training and testing datasets. On the test set, the model predicted most sessions as 0 which means non-purchase with 91.56% classified as non-revenue sessions and 8.439% as purchasing sessions (Figure 4.3.1.2). This distribution is much lower than the ground truth of the dataset where the purchasing sessions are 15.5%. These reflect the model prioritizing precision over recall by adopting conservative decision boundary.

Logistic Regression model trained successfully

Top 10 Positive Coefficients:

	Feature	Coefficient
8	PageValues	1.51395
23	Month_Nov	0.645140
24	Month_Oct	0.209774
19	Month_Jul	0.189187
5	ProductRelated_Duration	0.136155
25	Month_Sep	0.118004
11	Browser	0.056602
0	Administrative	0.056497
14	Weekend	0.053162
2	Informational	0.038816

Top 10 Negative Coefficients:

	Feature	Coefficient
10	OperatingSystems	-0.060395
1	Administrative_Duration	-0.063164
6	BounceRates	-0.100821
16	VisitorType_Returning_Visitor	-0.203772
20	Month_June	-0.227456
21	Month_Mar	-0.398955
22	Month_May	-0.410666
17	Month_Dec	-0.500591
7	ExitRates	-0.798303
18	Month_Feb	-1.100634

Model Intercept:
[-1.99202858]

Figure 4.3.1.1

```
Prediction Distribution for Logistic Regression:
0    2235
1     206
Name: count, dtype: int64

Prediction Distribution in Percentage:
0    91.56836
1     8.439164
Name: proportion, dtype: float64
```

Figure 4.3.1.2

3.3.1.2 Decision Tree

The decision tree classifier is selected as second techniques because it can capture non-linear relationships and complex interactions between the features. The decision tree model recursively splits the feature into smaller subsets based on the values which form a tree-like structure of decision rules that are easy to interpret.

To prevent the model from becoming too complex, it is initialized with random_state at 42, max_depth at 10 and min_samples_split at 50. The parameters are chosen to control the complexity and reduce the risk of overfitting while still allowing the model to learn meaningful patterns. After the model initialized, it was trained using the same training set from a logistic regression model. Upon training, the model had a tree depth of 10 and it consisted of 137 leaf nodes based on Figure 4.3.1.3.

The trained decision tree model is then used to generate predictions. The predictions result of the test set predicted approximately 84.43% of sessions as non-purchase and 15.567% as purchase sessions based on Figure 4.3.1.4. This distribution closely reflects the ground truth distribution of the dataset. This shows that the decision tree model is more sensitive to the minority class.

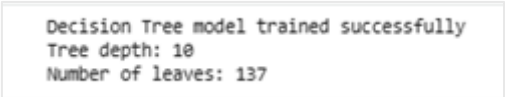


Figure 4.3.1.3

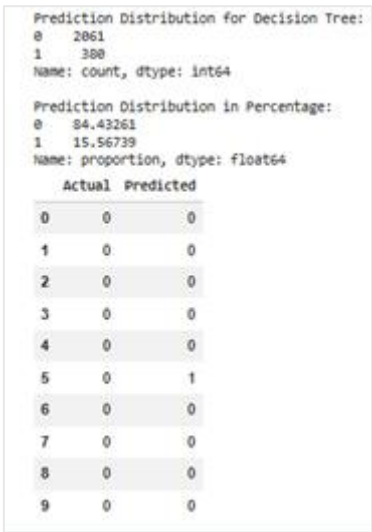


Figure 4.3.1.4

3.3.2 Association Rule Mining Technique

Association rule mining was applied to discover meaningful relationships between user behavioral patterns and purchase outcomes in online shopping sessions. Unlike classification techniques, which focus on prediction, association rule mining aims to uncover frequent co-occurring patterns that explain why certain sessions lead to revenue generation.

In this study, the Apriori algorithm was used to extract frequent itemsets and generate association rules from the dataset. Apriori works by identifying item combinations that occur together frequently based on a minimum support threshold and then generating rules that satisfy confidence and lift criteria.

To prepare the data for association rule mining, selected numerical features such as PageValues, ProductRelated, BounceRates, and ExitRates were discretized into categorical ranges (e.g., low, medium, high). Categorical features such as VisitorType, Weekend, and the target variable Revenue were retained directly. This transformation converts session-level data into a transaction-like format suitable for frequent pattern mining.

Frequent itemsets were first generated using a predefined minimum support threshold to ensure that only patterns occurring in a sufficient proportion of sessions were considered. From these itemsets, association rules were derived with constraints on minimum confidence and minimum lift, ensuring that the resulting rules are both reliable and meaningful.

The extracted rules were further categorized based on their consequences, such as Revenue-related, Engagement-related, and No-Revenue-related patterns, to support clearer interpretation and business insight generation.

Total number of frequent itemsets: 734

	support	itemsets
91	0.746006	(Revenue_Label_Revenue_No, PageValues_Bin_Low)
113	0.650307	(Weekend_Label_Weekday, Revenue_Label_Revenue_No)
92	0.598034	(Weekend_Label_Weekday, PageValues_Bin_Low)
110	0.458419	(Revenue_Label_Revenue_No, ExitRates_Bin_High)
102	0.440721	(BounceRates_Bin_High, Revenue_Label_Revenue_No)
...	...	...
66	0.058664	(Revenue_Label_Revenue_Yes, ProductRelated_Dur...
82	0.058091	(PageValues_Bin_High, ExitRates_Bin_High)
86	0.055961	(PageValues_Bin_High, Weekend_Label_Weekend)
26	0.053748	(ProductRelated_Duration_Bin_Short, ProductRel...
36	0.053585	(ProductRelated_Bin_Medium, Revenue_Label_Reve...

100 rows × 2 columns

Figure 4.3.2.1 Workflow of Association Rule Mining Using Apriori

3.4 Evaluation

3.4.1 Classification Model Evaluation

The performance of both logistic regression and decision tree models is evaluated using multiples metrics including:

- Accuracy: Used to measure overall correctness of predictions.
- Precision: Used to determine the percentage of the number of correctly predicted purchase sessions to all the predicted purchase.
- Recall: Used to measure whether the model can correctly identify actual purchase sessions.
- F1-Score: Used to provide balanced evaluations by combining precision and recall into one metric.
- ROC-AUC: Used to assess whether the model can distinguish between purchasing and non-purchasing sessions at various thresholds.

3.4.1.1 Logistic Regression

In Figure 4.4.1.1, the logistic regression model achieved accuracy of 0.8894, which means that 88.94% of test sessions are classified correctly. It also achieved a precision score of 0.7718 which indicates that when the model predicts a purchase, the correct percentage of prediction is 77.18%. However, the model achieved the recall value of

0.4162, which means that the model identifies 41.62% of actual purchasing sessions correctly. Additionally, the F1-score of 0.5408 shows that the model's ability to correctly identify purchase sessions is moderate. The model also achieves ROC-AUC with the value of 0.9022, which demonstrates that the logistic regression has a strong ability to distinguish between purchasing and non-purchasing sessions despite the limitations.

In Figure 4.4.1.2, the confusion matrix shows that the model correctly classified 2012 non-purchasing sessions and 159 purchasing sessions. However, it incorrectly classified 223 purchasing sessions as non-purchasing. These results show that the prediction by the model is high in false negatives. The outcome indicates that logistic regression prioritizes minimizing false positives which leads to higher precision but lower recall.

Logistic Regression Performance
 Accuracy : 0.8894
 Precision : 0.7718
 Recall : 0.4162
 F1-score : 0.5408
 ROC-AUC : 0.9022

Figure 4.4.1.1



Figure 4.4.1.2

3.4.1.2 Decision Tree

In Figure 4.4.1.3, the decision tree model has achieved accuracy of 0.8902, which means that 89.02% of test sessions are classified correctly. It also achieved precision score of 0.6500 which indicates that when the model predicts a purchase, the correct percentage of prediction is 65%. The model also achieved the recall value of 0.6466, which means that the model identifies 64.66% of actual purchasing sessions correctly. Additionally, the F1-score of 0.6483 shows that the model's ability to correctly identify purchase sessions is moderate to strong. Finally, the model also achieves ROC-AUC with a value of 0.9080 which demonstrates that logistic regression has a strong ability to distinguish

between purchasing and non-purchasing sessions despite the limitations. Overall, the decision tree model has improved accuracy, recall, F1-score, and ROC-AUC compared to logistic regression.

In Figure 4.4.1.4, the confusion matrix shows that the decision tree model correctly classified 1926 non-purchasing sessions and 247 purchasing sessions. One significant improvement is that the false negatives are reduced to 135 which shows the model improved sensitivity to minority class. Despite the increase of false positives, it is still acceptable as identifying purchase sessions is more critical than avoiding incorrect positive predictions.

```
Decision Tree Performance
Accuracy : 0.8902
Precision : 0.6500
Recall : 0.6466
F1-score : 0.6483
ROC-AUC : 0.9080
```

Figure 4.4.1.3

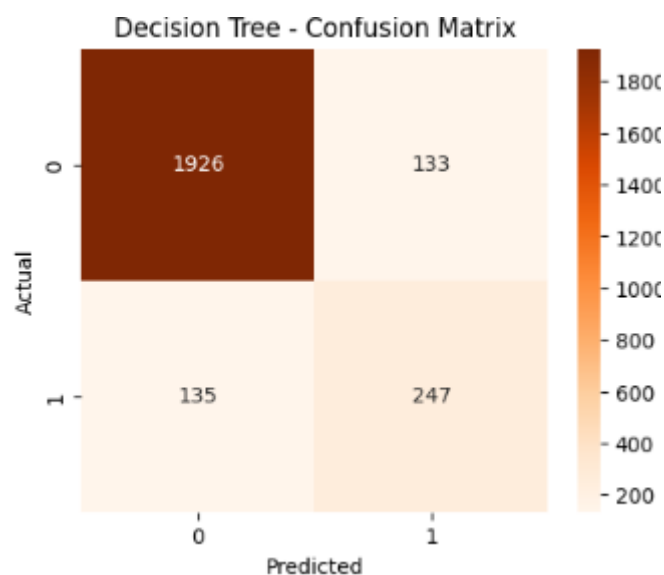


Figure 4.4.1.4

3.4.2 Association Rule Evaluation Measures

The quality and usefulness of association rules were evaluated using three standard metrics: Support, Confidence, and Lift.

Support: Measures how frequently a rule occurs in the dataset. It represents the proportion of sessions in which both the antecedent and consequent appear together. Rules with very low support were filtered out to avoid patterns that are statistically insignificant or overly specific.

Confidence: Indicates the reliability of a rule by measuring the conditional probability that the consequent occurs given that the antecedent has occurred. Higher confidence values imply stronger predictive relationships between shopping behaviors and outcomes.

Lift: Evaluates the strength of a rule relative to random chance. A lift value greater than 1 indicates a positive association between the antecedent and consequent, meaning that the presence of the antecedent increases the likelihood of the consequent occurring. Rules with higher lift values are considered more interesting and actionable, especially in marketing and decision-making contexts.

Total number of association rules: 2644					
	antecedents	consequents	support	confidence	lift
2448	ExitRates_Bin_Low, Weekend_Label_Weekday, Reve...	BounceRates_Bin_Low, PageValues_Bin_High	0.054568	0.631879	5.221447
1824	ExitRates_Bin_Low, Revenue_Label_Revenue_Yes	BounceRates_Bin_Low, PageValues_Bin_High	0.073822	0.624394	5.159597
1825	BounceRates_Bin_Low, PageValues_Bin_High	ExitRates_Bin_Low, Revenue_Label_Revenue_Yes	0.073822	0.610020	5.159597
2579	ProductRelated_Bin_High, Revenue_Label_Revenue...	ProductRelated_Duration_Bin_Long, PageValues_B...	0.051209	0.671321	4.897474
2580	ProductRelated_Duration_Bin_Long, Revenue_Labe...	ProductRelated_Bin_High, PageValues_Bin_High	0.051209	0.651721	4.850152
2447	Weekend_Label_Weekday, BounceRates_Bin_Low, Re...	ExitRates_Bin_Low, PageValues_Bin_High	0.054568	0.764638	4.617719
1826	BounceRates_Bin_Low, Revenue_Label_Revenue_Yes	ExitRates_Bin_Low, PageValues_Bin_High	0.073822	0.758418	4.580151
2611	ProductRelated_Duration_Bin_Medium, PageValues...	Revenue_Label_Revenue_Yes	0.050963	0.695749	4.450536
1820	ExitRates_Bin_Low, BounceRates_Bin_Low, PageVa...	Revenue_Label_Revenue_Yes	0.073822	0.680000	4.349790
2439	ProductRelated_Bin_High, Revenue_Label_Revenue...	ExitRates_Bin_Low, PageValues_Bin_High	0.054650	0.716434	4.326609
1667	BounceRates_Bin_Low, PageValues_Bin_High	Revenue_Label_Revenue_Yes	0.081770	0.675694	4.322246
2443	ExitRates_Bin_Low, Weekend_Label_Weekday, Boun...	Revenue_Label_Revenue_Yes	0.054568	0.671371	4.294593
2254	Weekend_Label_Weekday, BounceRates_Bin_Low, Pa...	Revenue_Label_Revenue_Yes	0.060549	0.664568	4.251078
2451	ProductRelated_Duration_Bin_Long, Revenue_Labe...	ExitRates_Bin_Low, PageValues_Bin_High	0.054568	0.694473	4.193987
1130	Revenue_Label_Revenue_Yes	ExitRates_Bin_Low, PageValues_Bin_High	0.101434	0.648847	3.918445
1127	ExitRates_Bin_Low, PageValues_Bin_High	Revenue_Label_Revenue_Yes	0.101434	0.612568	3.918445
2442	ExitRates_Bin_Low, Weekend_Label_Weekday, Boun...	PageValues_Bin_High	0.054568	0.876316	3.917741
1821	ExitRates_Bin_Low, BounceRates_Bin_Low, Revenu...	PageValues_Bin_High	0.073822	0.875607	3.914574
2612	ProductRelated_Duration_Bin_Medium, Revenue_La...	PageValues_Bin_High	0.050963	0.868715	3.883761
1827	ExitRates_Bin_Low, Weekend_Label_Weekday, Page...	Revenue_Label_Revenue_Yes	0.073822	0.605104	3.870700
	antecedents	consequents	support	confidence	lift

Figure 4.4.2.1 Evaluation Metrics of Generated Association Rules

In this study, thresholds were applied to retain only rules with:

- Moderate to high support
- Confidence above practical relevance levels
- Lift values greater than 1

These evaluation measures ensure that the selected rules are not only frequent but also meaningful and impactful for understanding shopper behavior.

3.5 Results and Discussion

3.5.1 Classification Results and Discussion

In Figure 4.5.1, it shows a direct comparison of the metrics by the two models we implemented. Logistic regression model shows a higher precision compared to decision tree model, but it has much lower accuracy, recall, F1-score and ROC-AUC. Despite the difference in the metrics, their ROC-AUC values indicate that both models have strong classification capabilities with decision trees achieving higher scores. In Figure 4.5.2 shows ROC curve comparison between logistic regression and decision tree. It highlights that the decision tree consistently achieved a higher true positive rate across the wide range of false positive rates. This suggests that decision tree model outperforms logistic regression in identification of purchase sessions.

In summary, the evaluation results show that the decision tree model is a more suitable model for the objectives of this study. This is because it shows higher accuracy, recall, F1-score and ROC-AUC which make it more aligned with business goal related to purchase intention prediction as it provides more practical value.

	Metric	Logistic Regression	Decision Tree	Difference (LR - DT)	Better Model	% Difference (LR vs DT)
0	Accuracy	0.8894	0.8902	-0.0008	Decision Tree	-0.09
1	Precision	0.7718	0.6500	0.1218	Logistic Regression	18.74
2	Recall	0.4162	0.6466	-0.2304	Decision Tree	-35.63
3	F1-Score	0.5408	0.6483	-0.1075	Decision Tree	-16.58
4	ROC-AUC	0.9022	0.9080	-0.0058	Decision Tree	-0.64

Figure 4.5.1

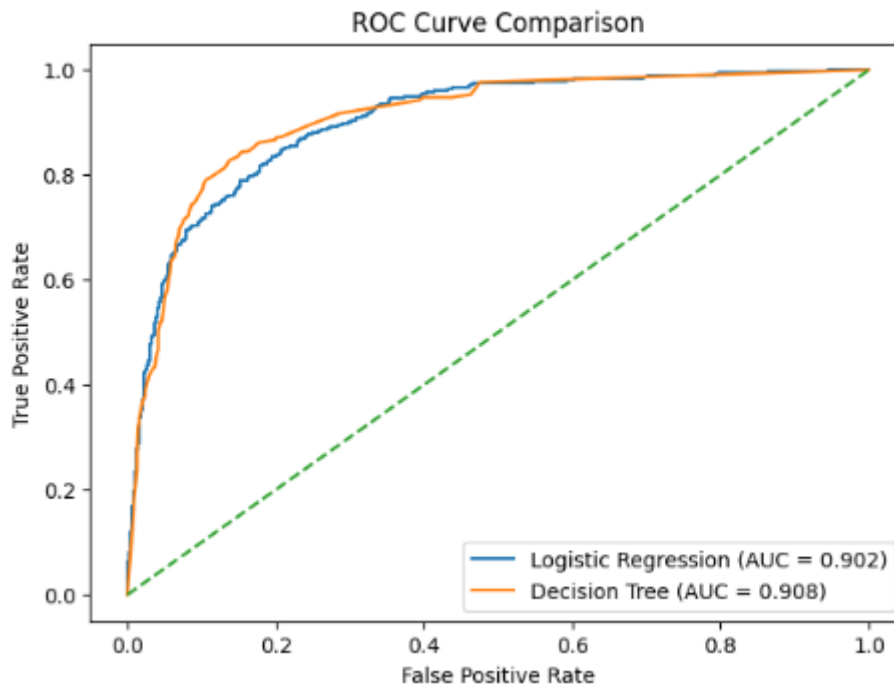


Figure 4.5.2

3.5.2 Association Rule Mining Results and Discussion

The association rule mining results reveal several important behavioral patterns related to online purchasing intentions. A significant number of high-quality rules were strongly associated with Revenue = True, highlighting key combinations of user engagement metrics that lead to successful conversions.

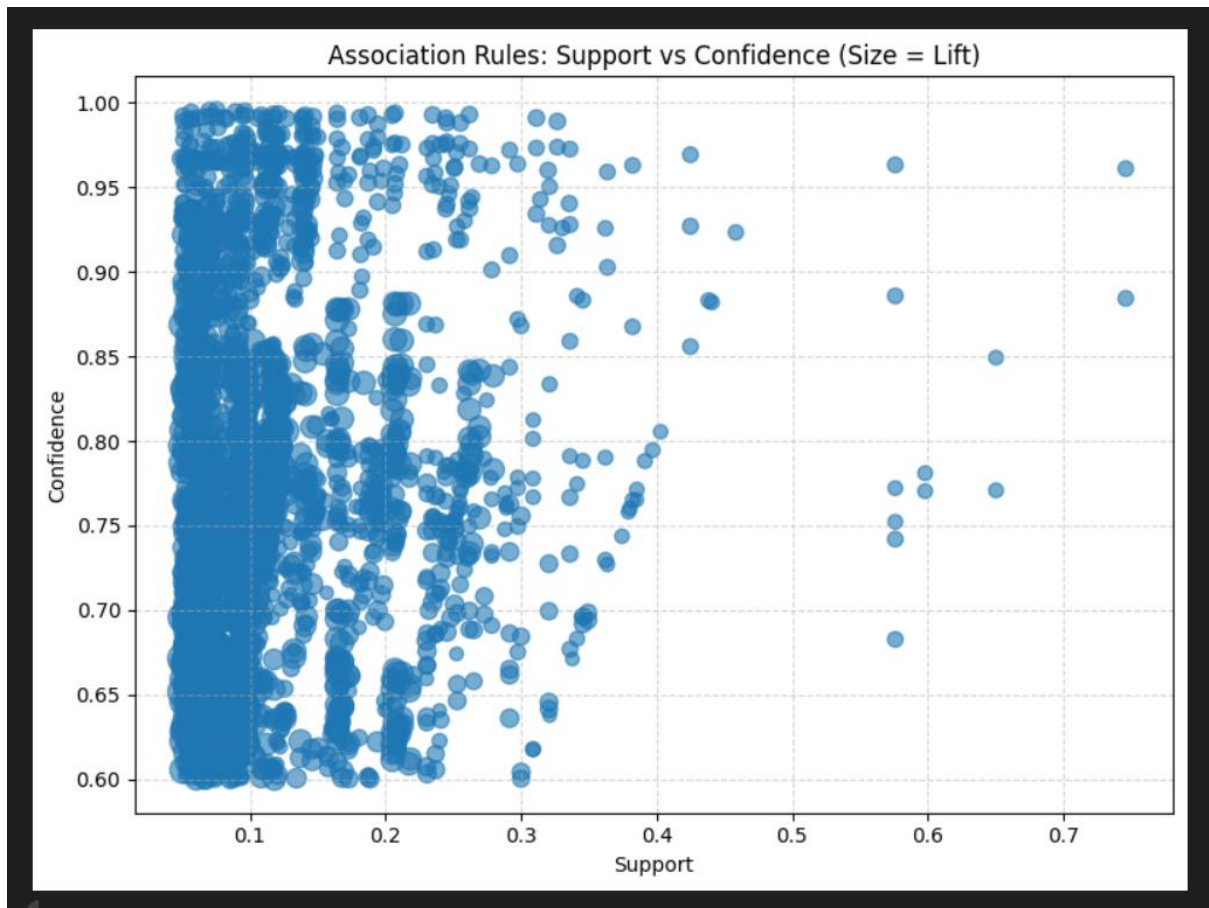


Figure 4.6.2.1 Visualization of Association Rules (Support vs Confidence)

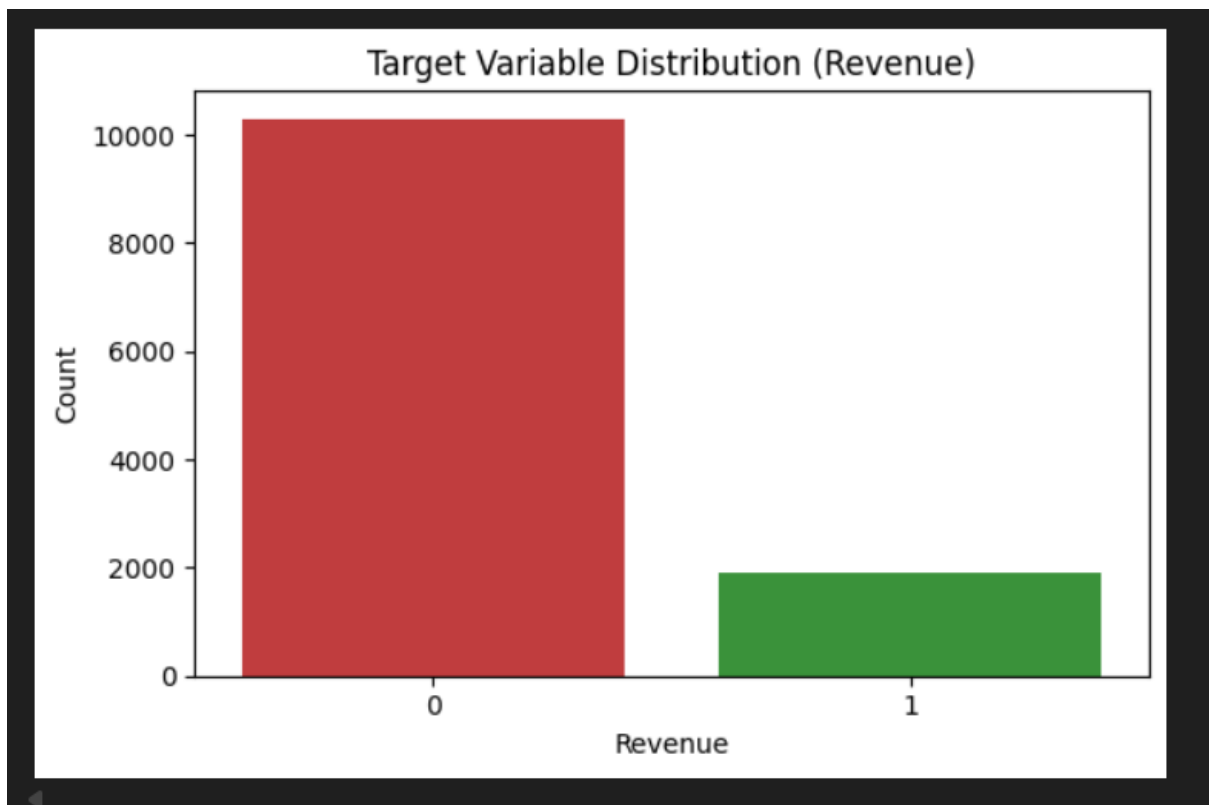


Figure 4.5.2.2 Distribution of Target Variable (Revenue)

One of the most prominent findings is the strong association between high PageValues and purchase outcomes. Sessions characterized by high page value consistently appeared in rules with high confidence and lift toward revenue generation, indicating that page value is a critical indicator of purchasing intent.

Another notable pattern involves Returning Visitors who exhibit strong engagement behavior, such as visiting multiple product-related pages and spending longer durations on those pages. These sessions frequently resulted in revenue-related rules with lift values significantly greater than 1, suggesting that engaged returning users are more likely to complete purchases.

Conversely, association rules linked to high bounce rates and exit rates were often associated with No Revenue outcomes. These patterns indicate that users who leave the site quickly or interact with fewer pages are less likely to convert, reinforcing findings observed during exploratory data analysis.

Temporal and contextual factors also emerged in several rules. Sessions occurring during special shopping periods or on weekends, when combined with strong engagement metrics, showed an increased likelihood of revenue generation. This suggests that timing, together with user behavior, plays an important role in influencing purchasing decisions.

3.6 Application

3.6.1 Application Overview

- **SPARK (Shopper Patterns and Analytics Research Kit)** is a Streamlit web application that presents the analysis results in an interactive dashboard. It allows users to explore model performance, run predictions with the trained models, browse association rules, and visualize the dataset.
- **Purpose:** Transform the Jupyter notebook analysis into an accessible web interface for viewing results, testing predictions, and exploring patterns.

3.6.2 Application Structure

The application uses **sidebar navigation** with five main sections:

Order	Section	Description
1	Overview	Project introduction, dataset summary (12,330 sessions), revenue and visitor-type charts, navigation guide.

2	Data Explorer	Interactive distributions, correlation heatmap, target analysis, sample data table with CSV download.
3	Model Performance	Comparison table (Accuracy, Precision, Recall, F1-Score, ROC-AUC), bar chart, confusion matrices for Logistic Regression and Decision Tree, winner summary.
4	Model Prediction	Input form (page visits, durations, bounce/exit rates, month, visitor type, etc.), real-time predictions from both models, probability scores, agreement indicator.
5	Association Rules	Rules loaded from trained model, filters (support 5–15%, confidence 60–100%, lift 1–6), top 10/20/30/50 rules table, scatter plot (support vs confidence, size = lift).

3.6.3 Technology Stack

- Framework: Streamlit
- Data: Pandas, NumPy
- Models: Scikit-learn (loaded from saved .pkl files)
- Visualization: Plotly
- Association rules: Exported from notebook (association_rules.csv , frequent_itemsets.csv)

3.6.4 File Structure

```
DataMiningProj/
|—— app.py           # Main Streamlit application
|—— models/
|   |—— logistic_regression.pkl  # Trained LR model
|   |—— decision_tree.pkl      # Trained DT model
```

		scaler.pkl	# Feature scaler
		feature_names.pkl	# Feature names for prediction
		association_rules.csv	# Exported association rules
		frequent_itemsets.csv	# Exported frequent itemsets
		online_shoppers_intention.csv	# Dataset
		requirements.txt	# Dependencies

3.6.5 How to Run

1. Install dependencies: `` pip install -r requirements.txt``
2. Ensure the dataset ``online_shoppers_intention.csv`` and the ``models/`` folder (with exported models and rules) are in the project directory.
3. Run the app: `` streamlit run app.py``
4. Open the URL shown in the terminal (e.g. ``http://localhost:8501``) in a browser.

3.6.6 Application Screenshots and Features

3.6.6.1 Overview Page

Feature	Description	Figure
Page title and intro	Project title, objectives, and short description of the dashboard.	Figure 4.7.6.1.1 — Overview: Introduction and Summary metrics
Summary statistics	Total sessions (12,330), revenue vs non-revenue counts and percentages.	Figure 4.7.6.1.1 — Overview: Introduction and Summary metrics
Revenue distribution chart	Bar or pie chart showing revenue vs no revenue (e.g. 84.5% / 15.5%).	Figure 4.7.6.1.2 — Overview: Revenue distribution and Visitor types
Visitor types chart	Bar chart of Returning Visitor vs New Visitor (Other excluded).	Figure 4.7.6.1.2 — Overview: Revenue distribution and Visitor types
Get Started / navigation guide	Bullet list of sections: Data Explorer, Model Performance, Model Prediction, Association Rules.	Figure 4.7.6.1.3 — Overview: Navigation guide

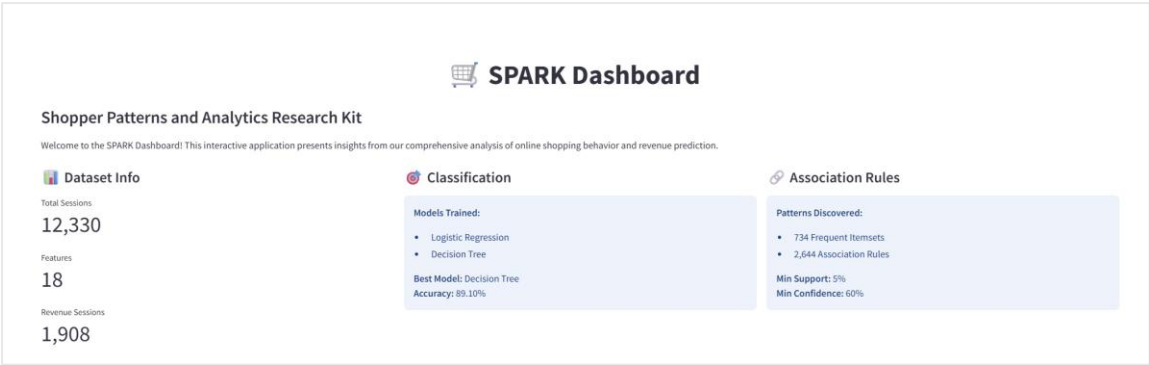


Figure 4.7.6.1.1 —Overview: Introduction and Summary metrics

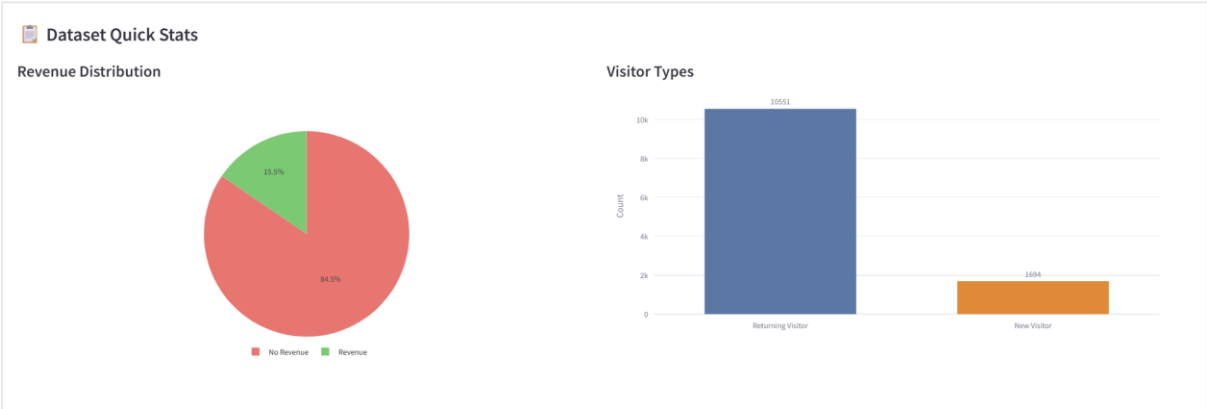


Figure 4.7.6.1.2 — Overview: Revenue distribution and Visitor types

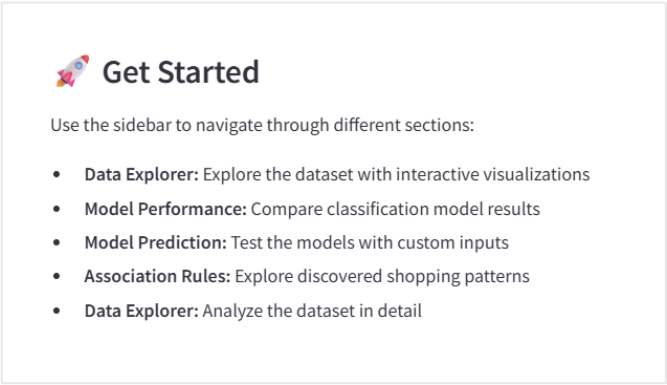


Figure 4.7.6.1.1 — Overview: Navigation guide

3.6.6.2 Data Explorer Page

Feature	Description	Figure
Page header	“Dataset Explorer” and short	Figure 4.7.6.2.1 — Data Explorer: Header,

	description of the dataset.	Distributions, Correlation heatmap
Data distribution visualizations	Interactive charts for key variables (e.g. page types, durations, rates).	Figure 4.7.6.2.1 — Data Explorer: Header, Distributions, Correlation heatmap
Correlation heatmap	Heatmap of feature correlations.	Figure 4.7.6.2.1 — Data Explorer: Header, Distributions, Correlation heatmap
Sample data section	“Sample Data” heading and slider for number of rows (e.g. 5–50).	Figure 4.7.6.2.2 — Data Explorer: Sample data controls, Sample data table, CSV download
Sample data table	Table showing first N rows of the dataset (real data).	Figure 4.7.6.2.2 — Data Explorer: Sample data controls, Sample data table, CSV download
Download option	Button to download dataset as CSV.	Figure 4.7.6.2.2 — Data Explorer: Sample data controls, Sample data table, CSV download

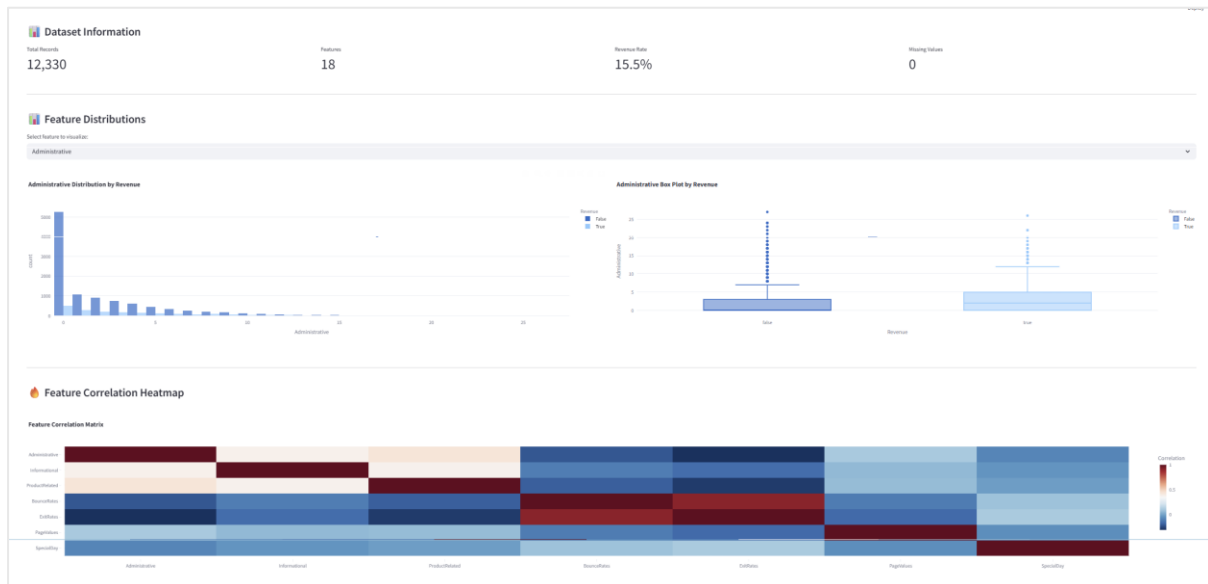


Figure 4.7.6.2.1 — Data Explorer: Header, Distributions, Correlation heatmap

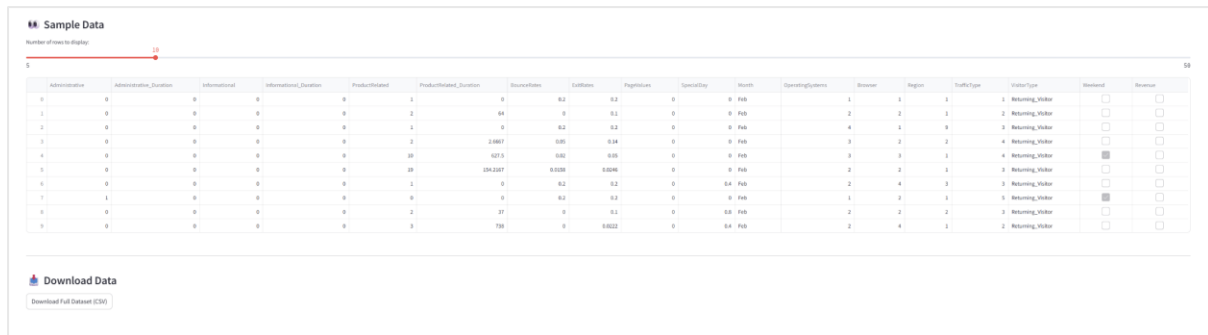


Figure 4.7.6.2.2 — Data Explorer: Sample data controls, Sample data table, CSV download

3.6.6.3 Model Performance Page

Feature	Description	Figure
Page header	“Model Performance Comparison” and short description of the models.	Figure 4.7.6.3.1 — Model Performance: Header, Metrics table, Bar chart, Winner
Performance metrics table	Table: Accuracy, Precision, Recall, F1-Score, ROC-AUC for Logistic Regression and Decision Tree (with highlighting for best per metric).	Figure 4.7.6.3.1 — Model Performance: Header, Metrics table, Bar chart, Winner
Metrics bar chart	Grouped bar chart comparing both models across metrics.	Figure 4.7.6.3.1 — Model Performance: Header, Metrics table, Bar chart, Winner
Winner announcement	Text stating winner (e.g. Decision Tree) with short justification.	Figure 4.7.6.3.1 — Model Performance: Header, Metrics table, Bar chart, Winner
Confusion matrix — Logistic Regression	Confusion matrix (TP, TN, FP, FN) for LR.	Figure 4.7.6.3.2 — Model Performance: LR confusion matrix, DT confusion matrix
Confusion matrix — Decision Tree	Confusion matrix for DT.	Figure 4.7.6.3.2 — Model Performance: LR confusion matrix, DT confusion matrix

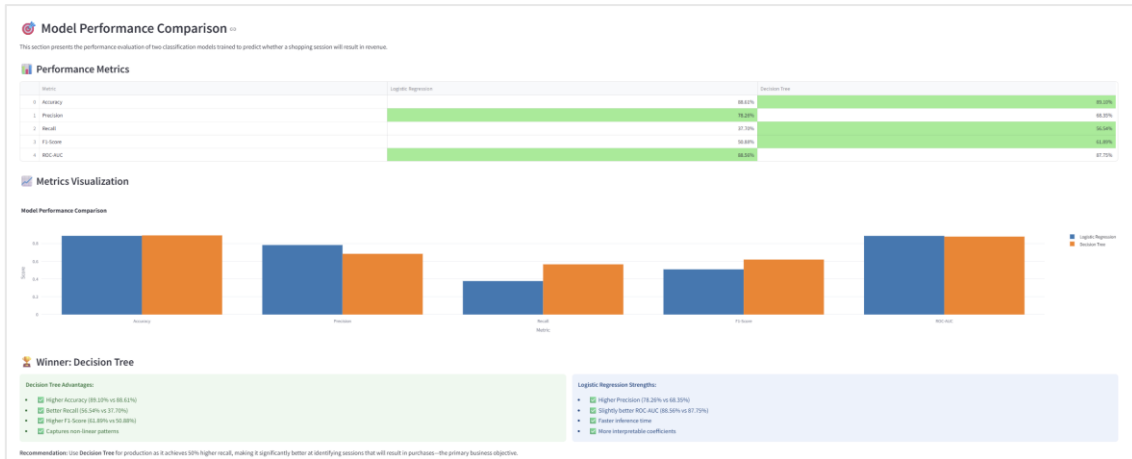


Figure 4.7.6.3.1 — Model Performance: Header, Metrics table, Bar chart, Winner

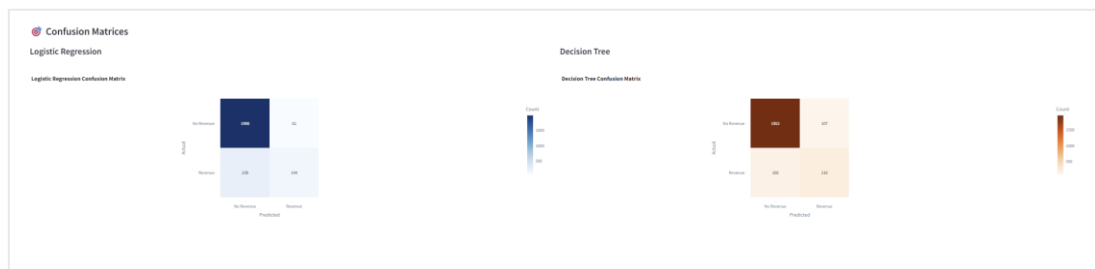


Figure 4.7.6.3.2 — Model Performance: LR confusion matrix, DT confusion matrix

3.6.6.4 Model Prediction Page

Feature	Description	Figure
Page header	“Revenue Prediction” and description of the prediction demo.	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement
Session features — Page visits	Sliders/inputs: Administrative, Informational, Product Related pages.	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement
Session features — Durations	Sliders: Administrative, Informational, Product Related durations (seconds).	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement
Session features — Engagement	Inputs: Bounce rate, Exit rate, Page values.	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement

Session features — Context	Month, Visitor type, Special day, Weekend (checkbox).	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement
Logistic Regression result	Predicted class (Revenue / No Revenue), probability, confidence level.	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement
Decision Tree result	Predicted class, probability, confidence level.	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement
Model agreement	Indicator showing whether both models agree on the prediction.	Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement

Revenue Prediction

Test the trained models by adjusting the session features below. The models will predict whether the session will result in revenue.

☒ Models loaded successfully

Enter Session Features

Note: Some technical features (Browser, OS, Region, Traffic Type) are automatically set to typical values for simplicity.

Page Visits	Duration (seconds)	Engagement Metrics
Administrative Pages: 0	Administrative Duration: 0	Source Rate: 0.00
Informational Pages: 0	Informational Duration: 0	Exit Rate: 0.00
Product Related Pages: 0	Product Related Duration: 0	Page Values: 0.00

Month: Feb Special Day: ☒ Weekend Session: ☐

Visitor Type: Returning_Visitor

Figure 4.7.6.4.1 — Model Prediction: Header, Session features, LR result, DT result, Agreement

3.6.6.5 Association Rules Page

Feature	Description	Figure
Page header	“Association Rules Discovery” and short description of rule mining.	Figure 4.7.6.5.1 — Association Rules: Header, Filters, Rules table
Filter controls	Sliders: Min Support (5–15%), Min Confidence (60–100%), Min Lift (1–6);	Figure 4.7.6.5.1 — Association Rules: Header, Filters, Rules table

	“Show Top” dropdown (10/20/30/50).	
Rules table	Table columns: #, Antecedents, Consequents, Support, Confidence, Lift, Category (e.g. Revenue / Engagement / No Revenue).	Figure 4.7.6.5.1 — Association Rules: Header, Filters, Rules table
Rules visualization	Scatter plot: Support vs Confidence, point size = Lift, color = Category (with jitter if used).	Figure 4.7.6.5.2 — Association Rules: Scatter plot, Category breakdown
Category breakdown	Counts or list of rules by category (Revenue, Engagement, No Revenue).	Figure 4.7.6.5.2 — Association Rules: Scatter plot, Category breakdown

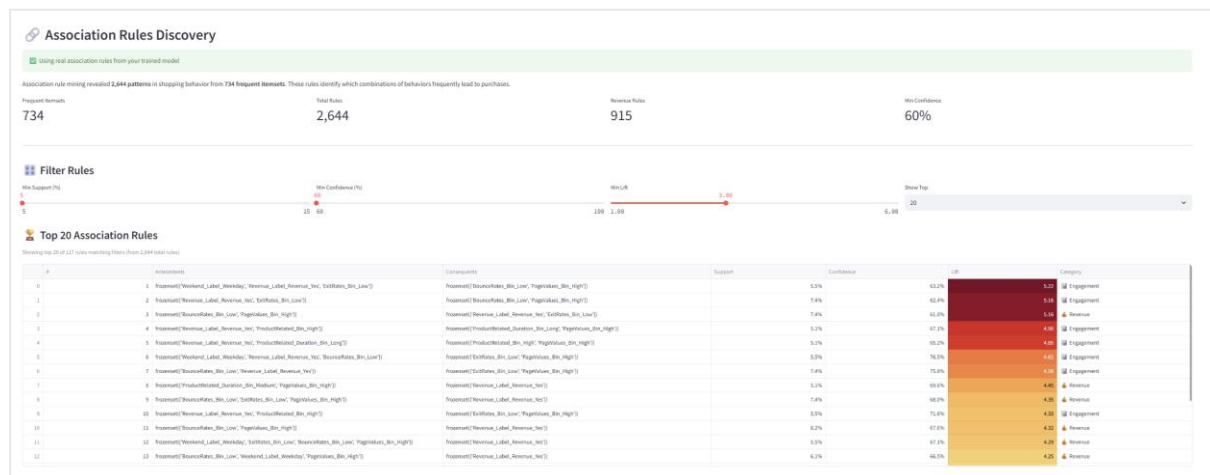


Figure 4.7.6.5.1 — Association Rules: Header, Filters, Rules table

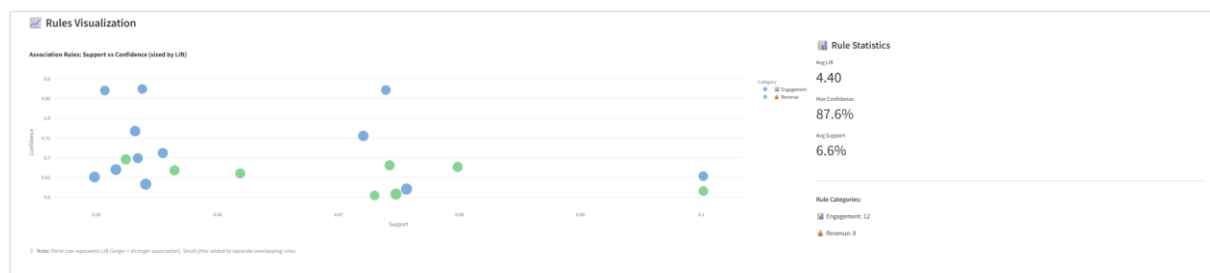


Figure 4.7.6.5.2 — Association Rules: Scatter plot, Category breakdown

3.6.6.6 Sidebar and Navigation

Feature	Description	Figure
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Sidebar title	“SPARK Dashboard” (or app title) in sidebar.	Figure 4.7.6.6.1 — Sidebar: Title
Navigation radio/list	Options: Overview, Data Explorer, Model Performance, Model Prediction, Association Rules.	Figure 4.7.6.6.1 — Sidebar: Navigation

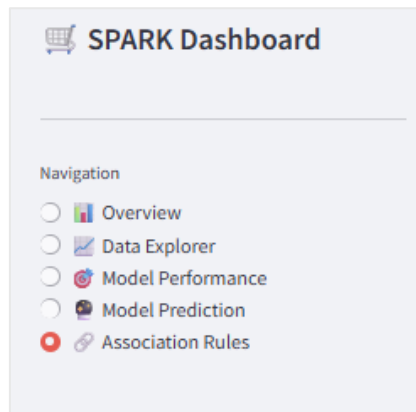


Figure 4.7.6.5.1 — Sidebar: Navigation

4 Conclusion and Future Work

4.1 Conclusion

The SPARK project proves that we can predict if a shopper will buy something just by looking at how they browse. Our main findings are:

- **Best Model:** The **Decision Tree** is the most useful model because it is much better at "catching" actual buyers than Logistic Regression.
- **Key Factors:** How much time a person spends on product pages and the "Page Value" are the biggest clues that they are going to buy something.
- **Shopper Habits:** New visitors have a higher chance of buying something compared to returning visitors.
- **Timing:** Shopping peaks in November, likely due to big sales events like Black Friday.
- **Tool Creation:** We successfully turned these complex results into an easy-to-use dashboard for business owners to use in real-time.

4.2 Future Work

To make the system even better, we plan to:

- **Better Tech:** Use more advanced AI, like Deep Learning, to understand even more complex shopping patterns.
- **Real-Time Updates:** Upgrade the system so it can learn and update itself instantly as new shoppers visit the site.
- **Long-Term Views:** Look at a shopper's history across many visits instead of just looking at one single session.
- **Live Testing:** Run "A/B tests" to see exactly how much these AI predictions help increase real sales.

References

Ahmed, I., Hoque, M. M., Banik, N., Rahman, A., Nur-E-Alam, M., & Islam, M. A. (2024). Optimizing classification methods for online buyers' purchase intentions in Bangladesh. *International Journal of Recent Technology and Engineering*, 12(6), 17–24. <https://doi.org/10.35940/ijrte.e7987.12060324>

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Sakar, C.O., Polat, S.O., Katircioglu, M. et al. *Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks*. *Neural Comput & Applic* 31, 6893–6908 (2019).

Appendix 1 Task Distribution

Member Name	Student ID	Technical Responsibilities	Report Responsibilities
Yong Di Lun	1211110326	• Design overall framework & pipeline • Implement Application	• Introduction (background, motivation, objectives)

		• Results & Discussion	• Application • Conclusion & Future Work
Low Sam Yee	1231303316	• Data Preprocessing	• Methodology – Overall Framework & Data Mining • Dataset section (source, attributes, volume) • Related Works • Data Preprocessing section
Hiew Wei Cheng	1211108075	• Classification. • Results & Discussion	• Implementation and Evaluation of Classification. • Results & Discussion
Nicky Kok Shee Ying	1221305657	• Association rule mining	• Implementation of association rule mining • Results & Discussion
All Members	—	• Model validation & testing • Result verification	• References (APA) • Appendix • Proofreading & formatting